

Dog Skin Disease Classification Using Deep Learning

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1. Project Title

Dog Skin Disease Classification Using Deep Learning Techniques

2. Introduction

Skin diseases are common health problems in dogs and can affect their quality of life. Early diagnosis is important for proper treatment. Manual diagnosis requires veterinary expertise and may take time. Deep learning techniques can assist in automatically identifying skin diseases from images, helping veterinarians and pet owners make faster decisions.

3. Problem Statement

The objective of this project is to develop a machine learning model that can classify different types of dog skin diseases using image data. The model should accurately identify disease categories from uploaded dog skin images.

4. Objectives

- Collect and analyze dog skin disease image data.
- Preprocess image datasets for model training.
- Develop deep learning models for classification.
- Compare model performance.
- Evaluate accuracy and prediction results.

5. Dataset

The dataset is obtained from Kaggle and contains labeled images of dogs with different skin disease conditions as well as healthy skin samples.

Dataset characteristics:

- Image-based dataset
- Multiple disease categories
- Labeled training samples
- Suitable for image classification tasks

6. Methodology

The following steps will be performed:

1. Data Collection
2. Data Preprocessing
3. Image Resizing and Normalization
4. Model Development
5. Model Training
6. Performance Evaluation
7. Result Analysis

7. Models Used

Custom Convolutional Neural Network (CNN)

A custom CNN model is developed to learn image features and classify dog skin diseases.

MobileNetV2

MobileNetV2 is a lightweight pre-trained deep learning architecture used for transfer learning and improved classification performance.

8. Evaluation Metrics

The following metrics will be used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

9. Expected Results

The project is expected to achieve reliable classification performance and demonstrate the effectiveness of deep learning techniques for dog skin disease detection.

10. Tools and Technologies

- Python
- TensorFlow
- Keras
- NumPy
- Matplotlib
- Scikit-learn
- Kaggle
- GitHub

11. Conclusion

This project aims to develop an intelligent image classification system for dog skin disease detection. The implementation of deep learning models such as CNN and MobileNetV2 can support accurate disease identification and contribute to improved veterinary diagnosis.